

Design for a Tape Spring Five-Bar Parallel Robot Arm System Incorporating Edge AI Computer Vision

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INTRODUCTION

Robotic arms are crucial to automation in industrial manufacturing, and are also used in many other fields. This project designed a self-contained robotic arm system which uses computer vision to control a (believed to be) novel type of arm. The arm consists of a single continuous tape spring with joints formed from 3D printed nodes that pinch the tape to allow pivoting. Because the motors are only on the systems base, with 3/5 joints being passive, the arm is lightweight, folds to a compact size, and is cheap to produce. This poster presents an overview of the system design and results from testing.

METHODS

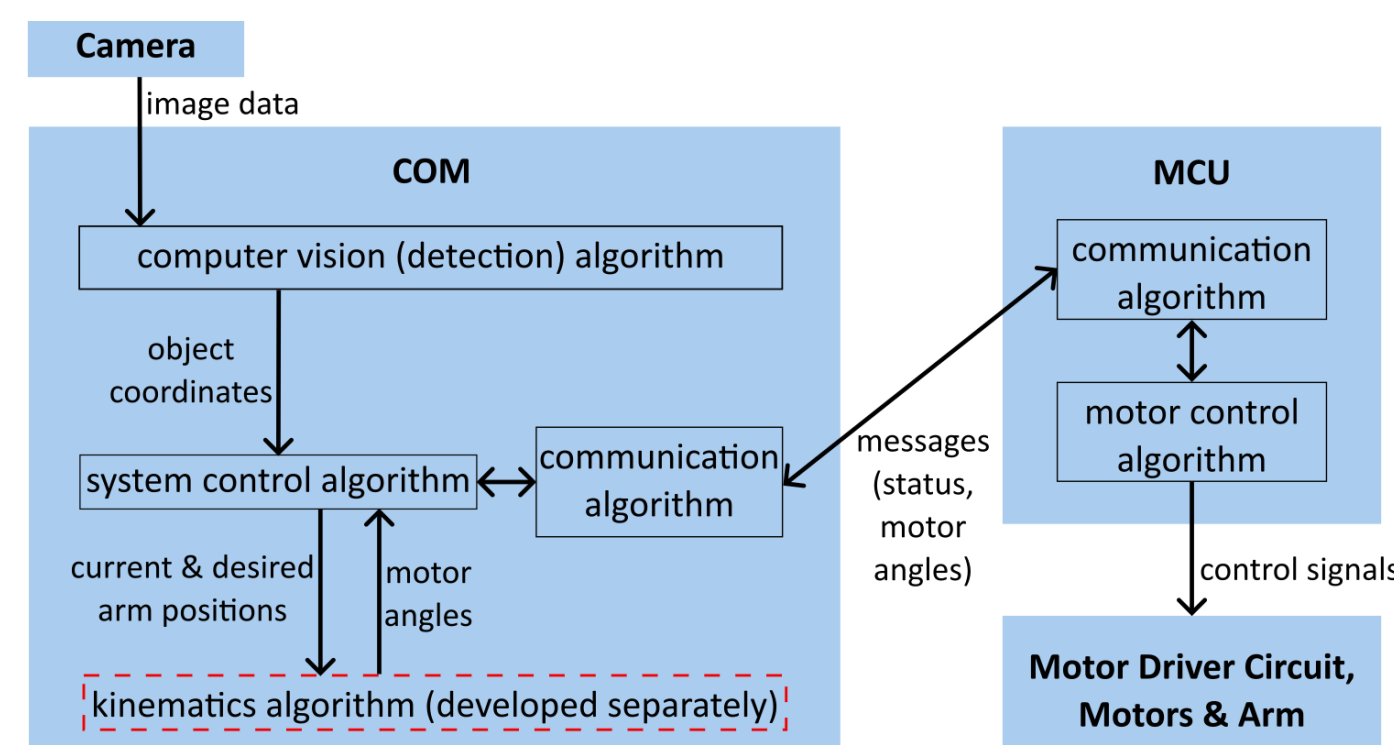


Figure 1: Overview of the system architecture.

The system uses a modular architecture for both hardware and software. Key components selected include an Nvidia Jetson Orin Nano COM (Computer On Module) as the main processing platform, a basic MCU (Microcontroller Unit), and stepper motors rather than the more expensive servo motors typically found on arms of a similar (but rigid) design [2]. The design prioritises maximising the arms inherent strengths, especially portability and low manufacturing costs, while managing its inherent challenges. Testing focused primarily on accuracy of the computer vision system and the arm movements.

RESULTS

Arm & Motors

Resonance and vibrations caused by the stepping nature of the motors, plus tension at joint nodes, initially resulted in skipped steps and significant accuracy loss. Identifying and avoiding resonant speeds, using small microsteps between the main step positions to minimise vibrations, and only pausing at full step positions to maximise torque resolved this, but resulted in an effective motor resolution of 1.8° . Motor resolution and arm configuration are the primary contributors to arm error (figure 4), with the current system only suitable for lower precision tasks. Mechanisms which allow the arm to change length autonomously, so it can optimise its configuration for a specific movement, will significantly improve accuracy and are currently being developed.

Computer Vision

Aruco markers [3] were found to be the optimal method for measuring the motor positions. A YOLO [1] (You Only Look Once) CNN (convolutional neural network) was trained from scratch on 4 objects. With some processing optimisations it was able to run at 36 frames per second on the COM, with 99.6% detection accuracy and 92.2% classification accuracy even with some video frames having objects partially occluded and in shadows. Further refinement of the CNN would make it suitable for tasks such as identifying parts to move in a warehouse.

As shown in figure 4, the camera accuracy is sufficient for low to medium precision applications. The primary contributor to camera error was insufficient perpendicularity with the measured surface, which could be mitigated by using binocular vision, though this would increase hardware costs.

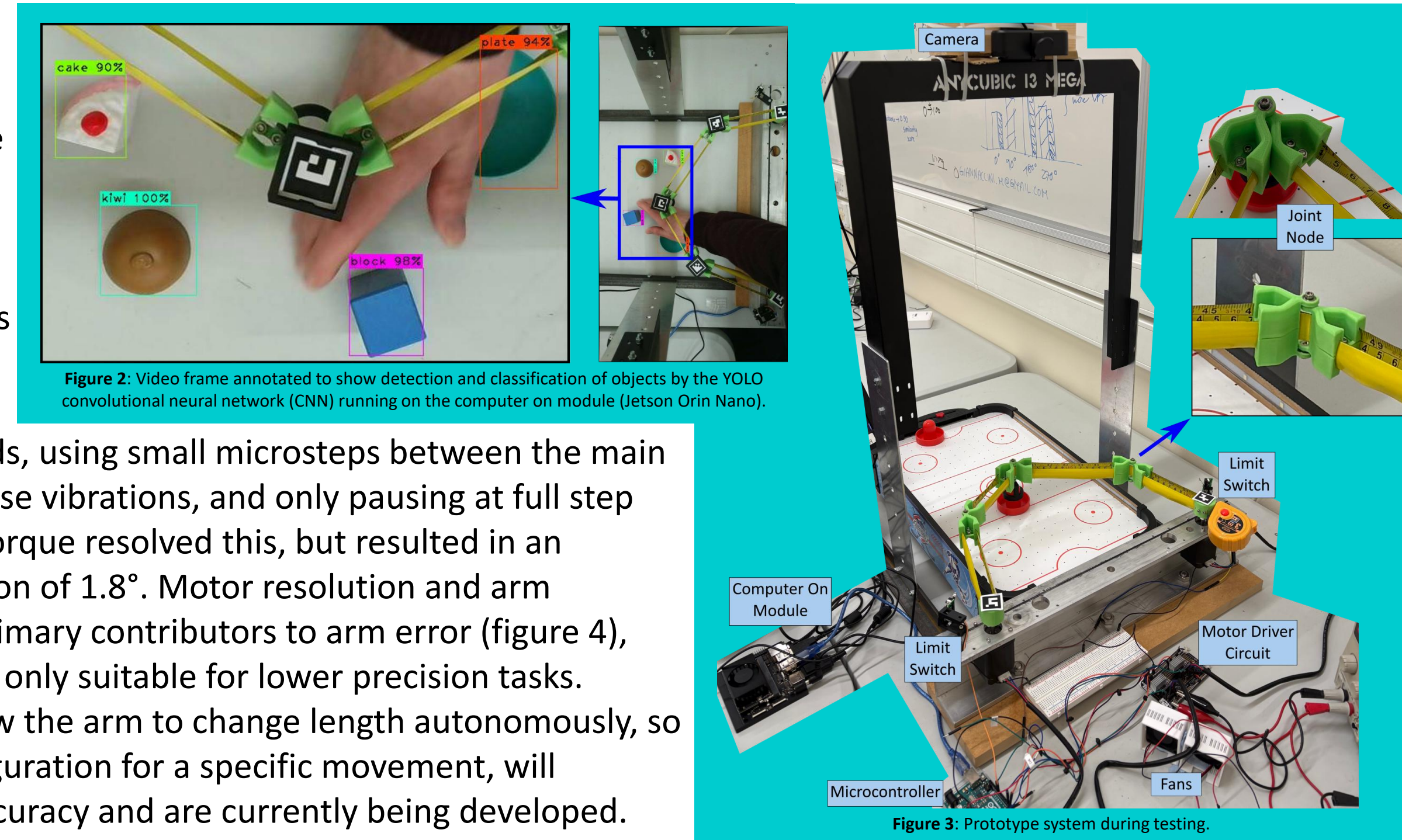


Figure 3: Prototype system during testing.

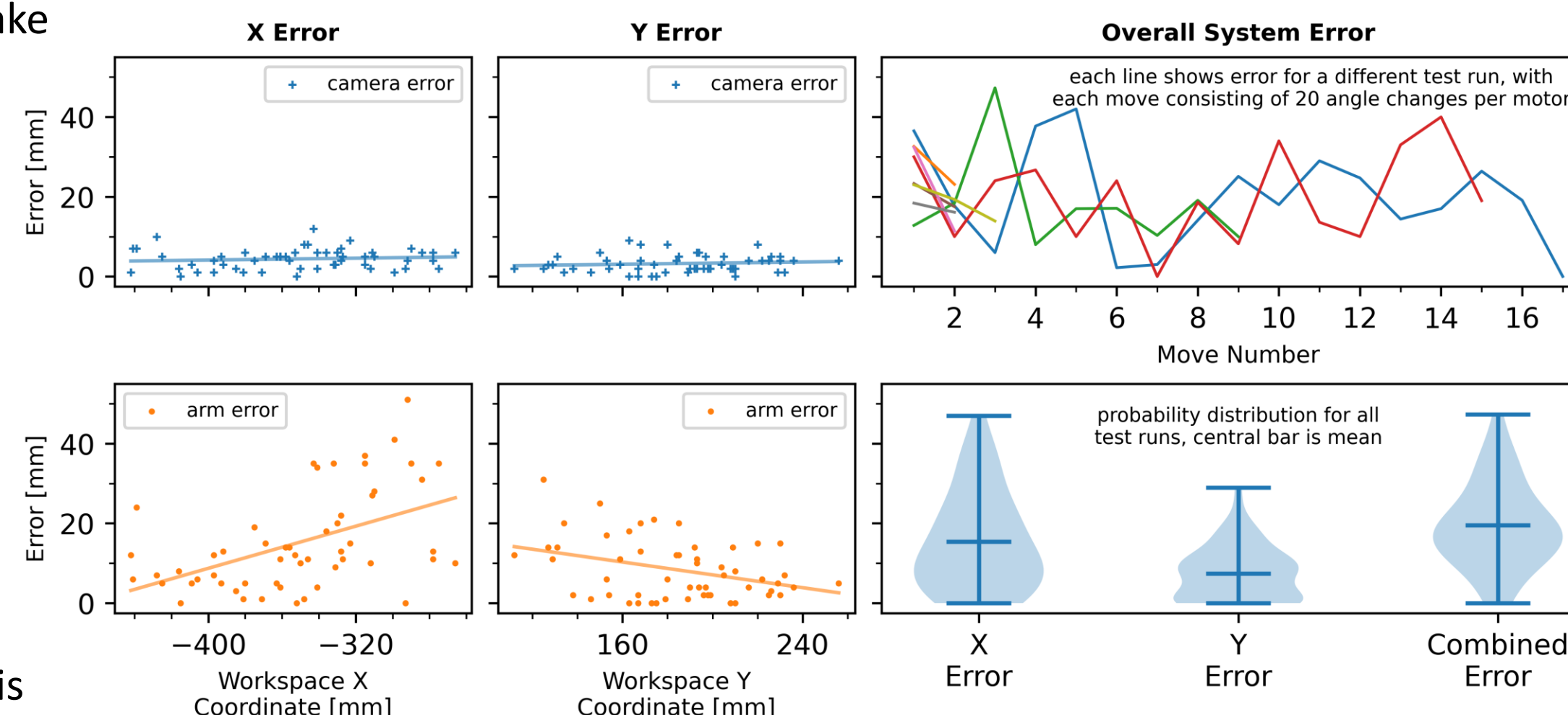


Figure 4: Summary of results from system accuracy testing.

DISCUSSION

The project demonstrated that this type of arm can be incorporated into a working system with low hardware costs. The use of a YOLO CNN enabled the COM to detect and classify objects at a high frame rate relative to its processing power. The mean arm accuracy of 20 mm makes it suitable for lower precision tasks, for example manipulating the camera of an inspection robot. Arm accuracy will improve if future work enables the arm bar lengths to be autonomously changed. The project identified two key areas where more expensive hardware could significantly increase system accuracy; changing the motors to servo motors, and adding a second camera to enable binocular vision. Future work may want to develop cheaper and more expensive higher performance system versions in tandem, which thanks to the modular architecture of the design could be done without significant additional overhead.

Highlights

- Prototype system for a new type of robotic arm designed, constructed and successfully tested.
- Key challenges of controlling the arm resolved.
- Options to upgrade performance identified and can be implemented on existing prototype.
- Demonstrated the effectiveness of YOLO CNNs for AI computer vision on embedded hardware.

CONCLUSIONS

This project has explored the challenges and benefits of the new type of arm, created a prototype system which can be used to further advance research towards a viable product, and identified potential improvements over and above those that were implemented during design and testing of the system.

Acknowledgements

- Sincere gratitude to my supervisor, Dr. Andres San Millan Rodriguez, for his guidance and support.
- Thanks to Zayne Turnbull for writing the kinematics algorithm and his collaboration on the wider arm project.

References

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